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Github link: https://github.com/Dhanushsj12/orbitdesk-support-agent

Video link: https://drive.google.com/file/d/11ISA15G7pOJnnbTSp-5B5bBD1WjeXnTU/view?usp=drivesdk

OrbitDesk AI Support Agent: A Retrieval-Augmented Generation (RAG) System for Intelligent Customer Support

ABSTRACT

Customer support teams often spend significant time answering repetitive questions related to product features, user permissions, API usage, and troubleshooting. While knowledge base articles contain the required information, searching through multiple documents manually can be time-consuming. This project aims to simplify that process by developing an AI-powered support assistant capable of retrieving relevant information and generating accurate responses.

The OrbitDesk AI Support Agent combines Retrieval-Augmented Generation (RAG) with a local Large Language Model to answer user queries based only on the available documentation. Instead of relying on general knowledge, the system searches a vector database containing knowledge base articles and resolved support cases, retrieves the most relevant documents, and generates responses supported by those documents.

The application also verifies generated answers before presenting them to the user and supports different workflows such as clarification requests, escalation handling, and out-of-scope detection. This approach improves response reliability while reducing the possibility of incorrect or unsupported answers.

1. INTRODUCTION

Artificial Intelligence has transformed the way organizations provide customer support. Traditional chatbot systems generally depend on predefined rules or keyword matching, which limits their ability to answer complex questions. Recent advancements in Large Language Models (LLMs) have improved conversational capabilities, but these models may generate responses that are inaccurate or unsupported by official documentation.

To overcome this limitation, Retrieval-Augmented Generation (RAG) combines document retrieval with language generation. Instead of relying solely on the language model's knowledge, the system first retrieves relevant documents and then generates responses based only on the retrieved information.

The OrbitDesk AI Support Agent follows this approach. It uses a local language model together with semantic document retrieval to answer technical support questions accurately while ensuring that responses remain grounded in official documentation.

2. PROBLEM STATEMENT

Customer support engineers frequently receive repeated questions regarding user permissions, API credentials, workspace settings, and troubleshooting procedures. Searching through documentation manually increases response time and may lead to inconsistent answers.

General-purpose AI models can answer such questions but often produce responses that are not supported by the organization's documentation. This creates a risk of misinformation, especially in enterprise support environments.

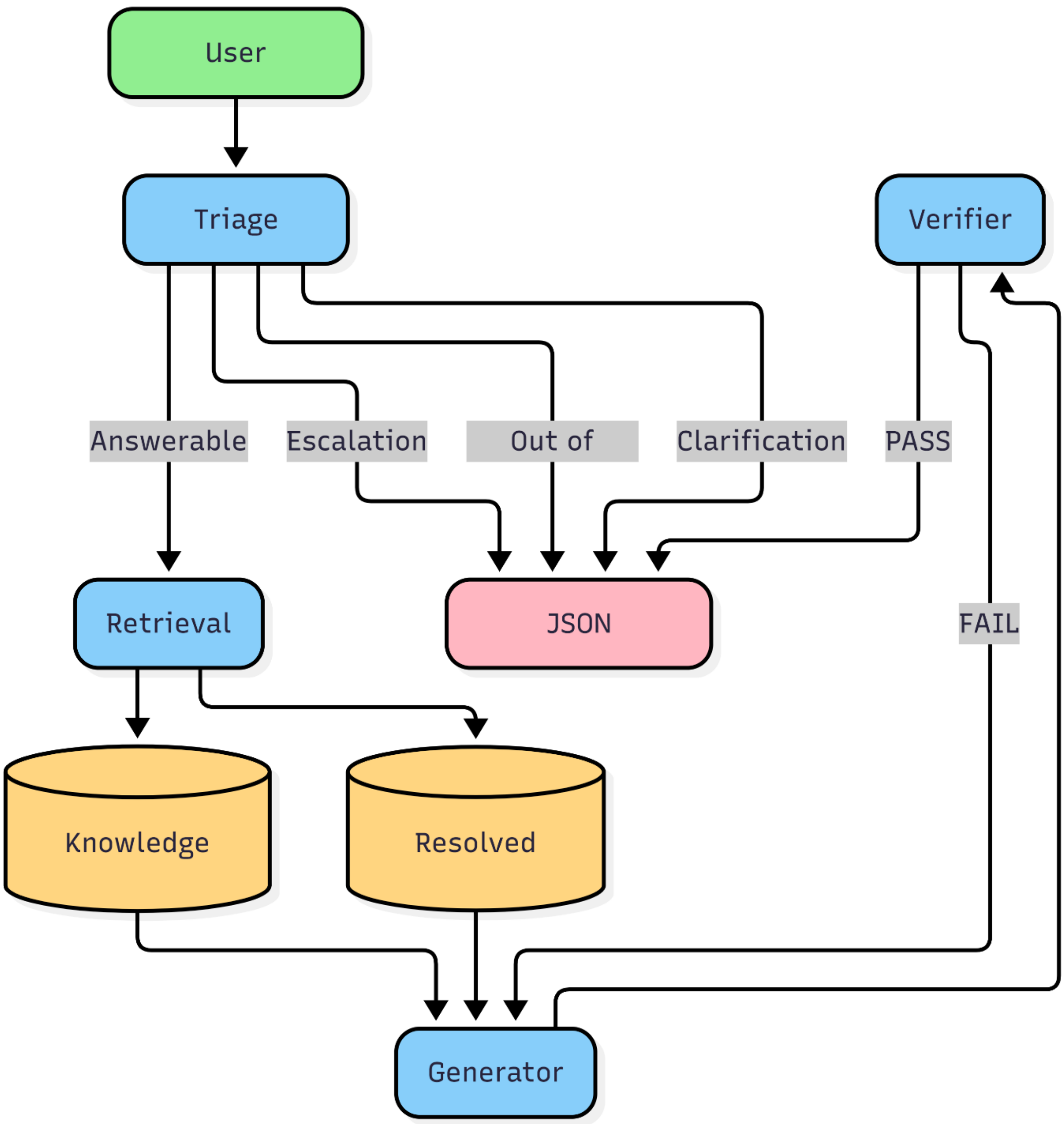
The objective of this project is to build an intelligent support assistant capable of retrieving relevant documentation, generating accurate responses, verifying those responses, and handling situations where escalation or clarification is required.

3. OBJECTIVES

The main objectives of this project are:

- Develop an AI-powered support assistant for OrbitDesk documentation.
- Implement Retrieval-Augmented Generation (RAG).
- Retrieve relevant knowledge base documents using semantic search.
- Generate responses using a local Large Language Model.
- Verify generated responses before displaying them.
- Detect questions that require clarification.
- Identify issues requiring human escalation.
- Detect requests outside the scope of available documentation.
- Return structured responses with confidence scores and supporting sources.

4. SYSTEM ARCHITECTURE



The OrbitDesk AI Support Agent follows a modular workflow implemented using LangGraph. The architecture is divided into four main processing stages: **Triage**, **Retrieval**, **Generation**, and **Verification**.

Initially, the user submits a support query to the system. The query is processed by the **Triage Node**, which determines the appropriate execution path. Based on predefined classification rules, the request is categorized as **Answerable**, **Requires Escalation**, **Requires Clarification**, or **Out of Scope**.

If the request is classified as **Answerable**, it is forwarded to the **Retrieval Node**. The Retrieval Node performs semantic similarity search over two data sources: the **Knowledge Base** and previously **Resolved Support Cases**. The most relevant documents are selected and passed to the Generator.

The **Generator Node** receives both the user query and the retrieved documents. Using the local language model, it generates a response that is grounded only in the retrieved evidence.

The generated response is then passed to the **Verifier Node**. The verifier checks whether the answer is supported by the retrieved documentation. If verification succeeds, the system returns the final structured JSON response. If verification fails, the answer generation process is retried until the configured retry limit is reached.

For requests classified as **Requires Escalation**, **Requires Clarification**, or **Out of Scope**, the system bypasses the retrieval and generation stages and directly returns an appropriate JSON response to the user.

This modular architecture improves maintainability, reduces unnecessary processing, and ensures that only verified, evidence-based responses are returned.

. TECHNOLOGIES USED

Technology	**Purpose**
Python	Application development
LangGraph	Workflow orchestration
FAISS	Vector similarity search
Sentence Transformers	Text embedding generation
TinyLlama	Local language model

Technology	Purpose
PyTorch	Model execution
Logging Module	Application logging

6. METHODOLOGY

The implementation follows a Retrieval-Augmented Generation pipeline.

Initially, knowledge base articles and resolved support cases are converted into vector embeddings using the Sentence Transformer model. These embeddings are stored in a FAISS vector index.

When a user submits a question, the triage component classifies the request. If the request is answerable, semantic similarity search retrieves the most relevant documents. The retrieved information is then supplied to the language model, which generates a response based solely on the available documentation.

Finally, the generated response undergoes verification before being returned to the user in a structured JSON format.

Hardware Configuration

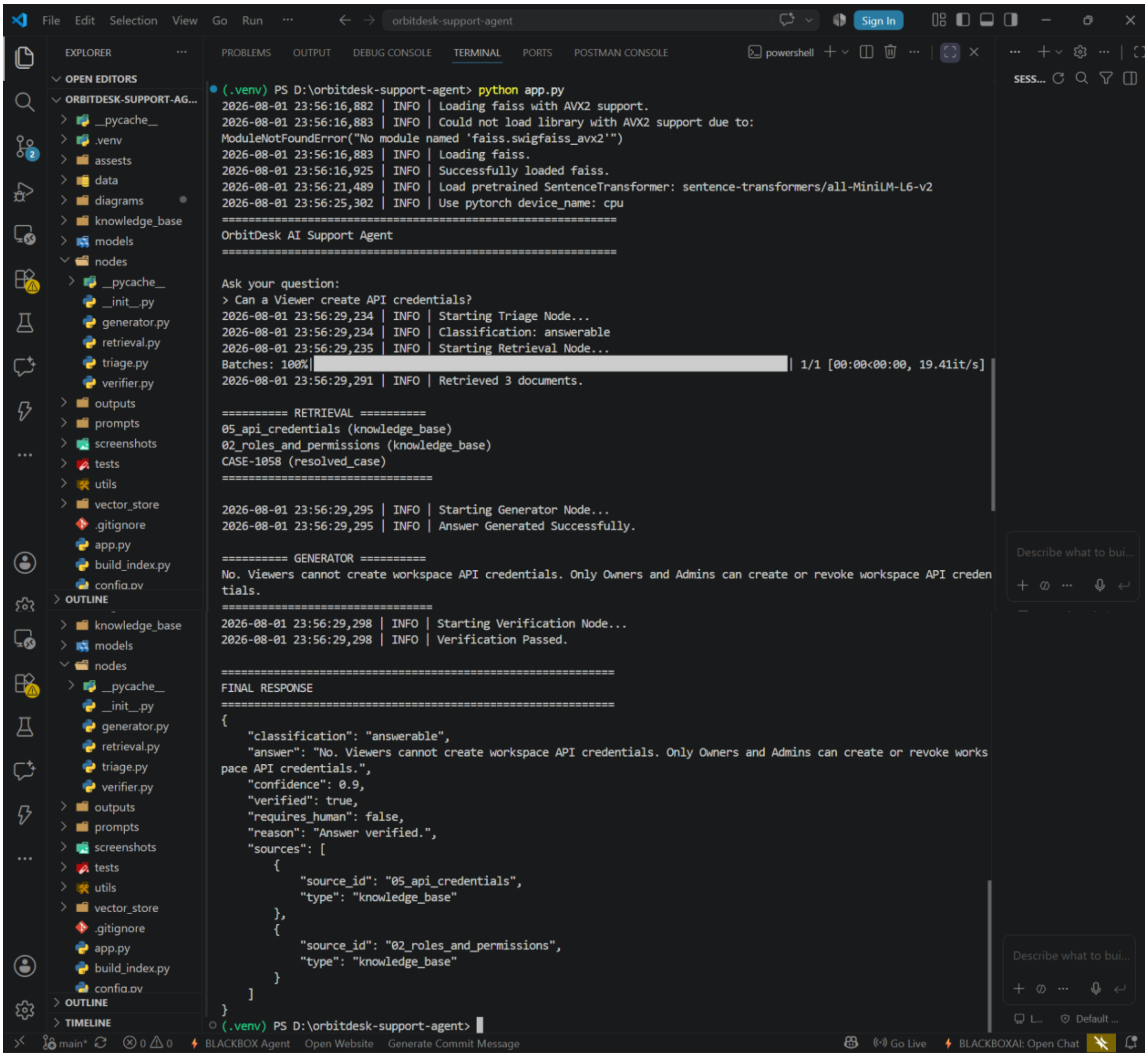
Component	Specification
Laptop	Samsung Galaxy Book2
Processor	**12th Gen Intel® Core™ i5-1235U (10 Cores, 12 Threads, Base Clock 1.30 GHz)**
RAM	**16 GB**
Storage	SSD
Graphics	Intel® Iris® Xe Graphics (Integrated)

Component	Specification
Operating System	Windows 11
Execution Device	CPU

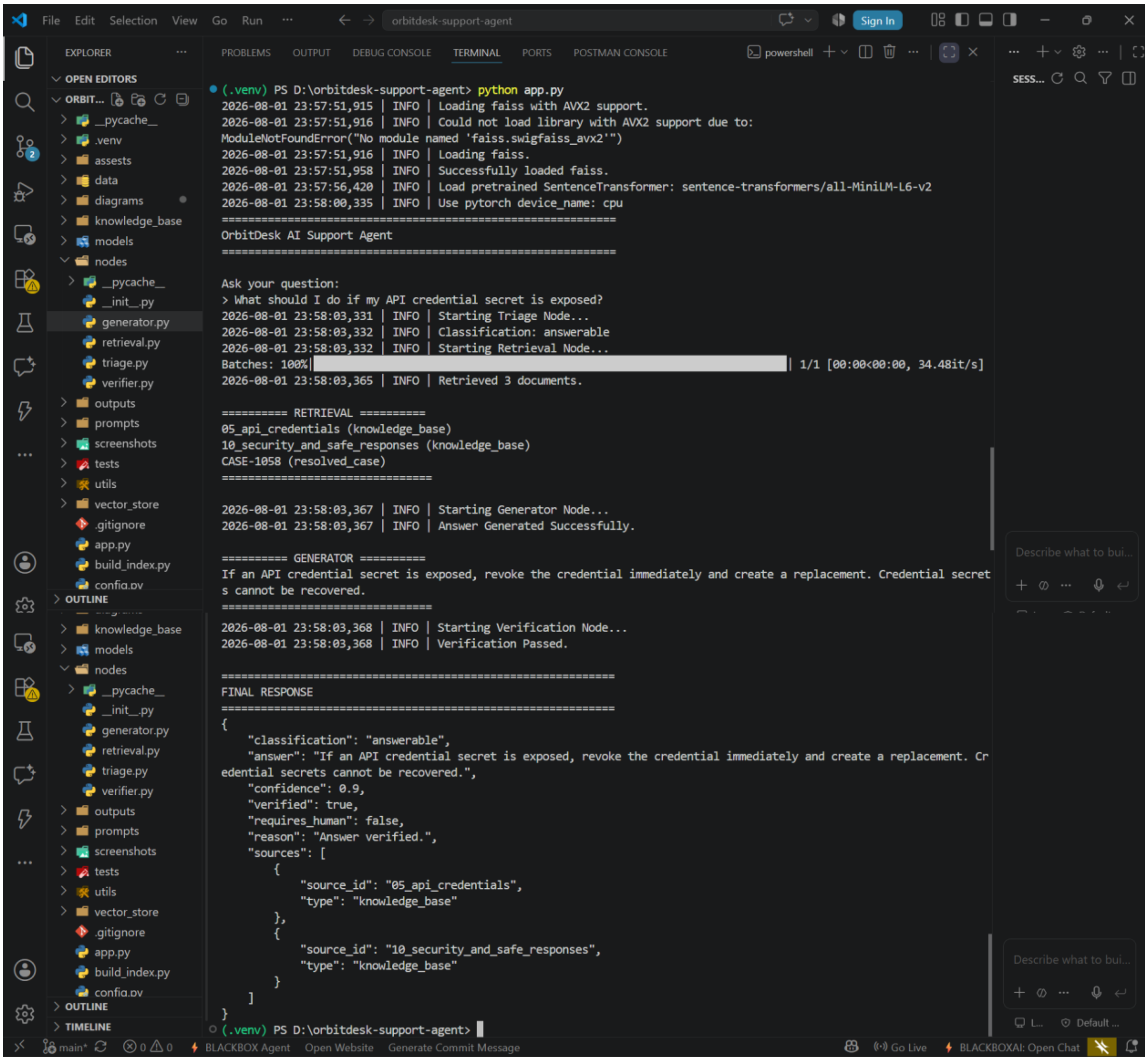
Software Configuration

Component	Specification
Programming Language	Python 3.11
IDE	Visual Studio Code
Workflow Framework	LangGraph
Embedding Model	sentence-transformers/all-MiniLM-L6-v2
Language Model	TinyLlama-1.1B-Chat-v1.0
Deep Learning Framework	PyTorch
Transformers Library	Version 4.57.6
Vector Database	FAISS (CPU)
Version Control	Git & GitHub

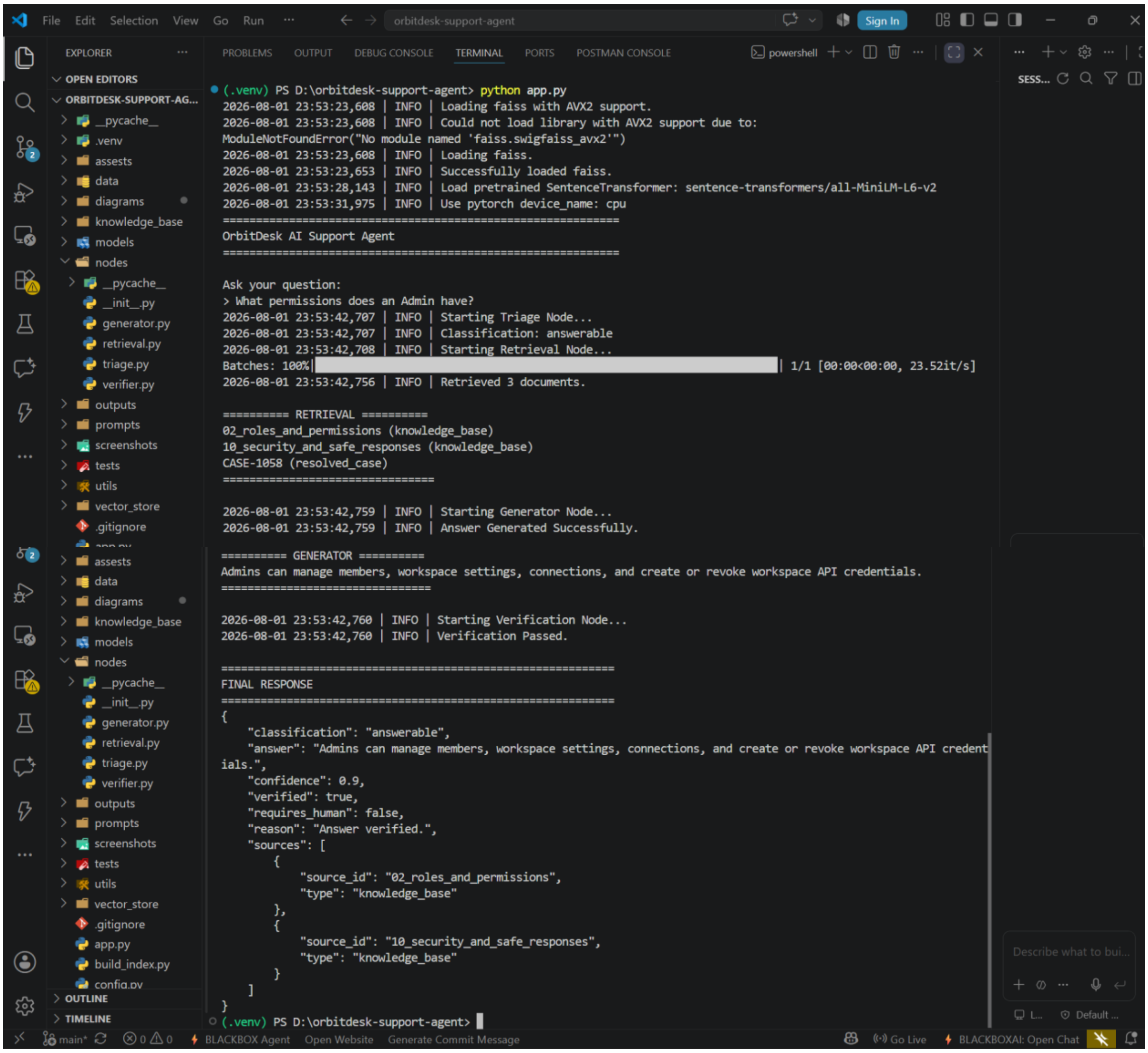
Screenshots:

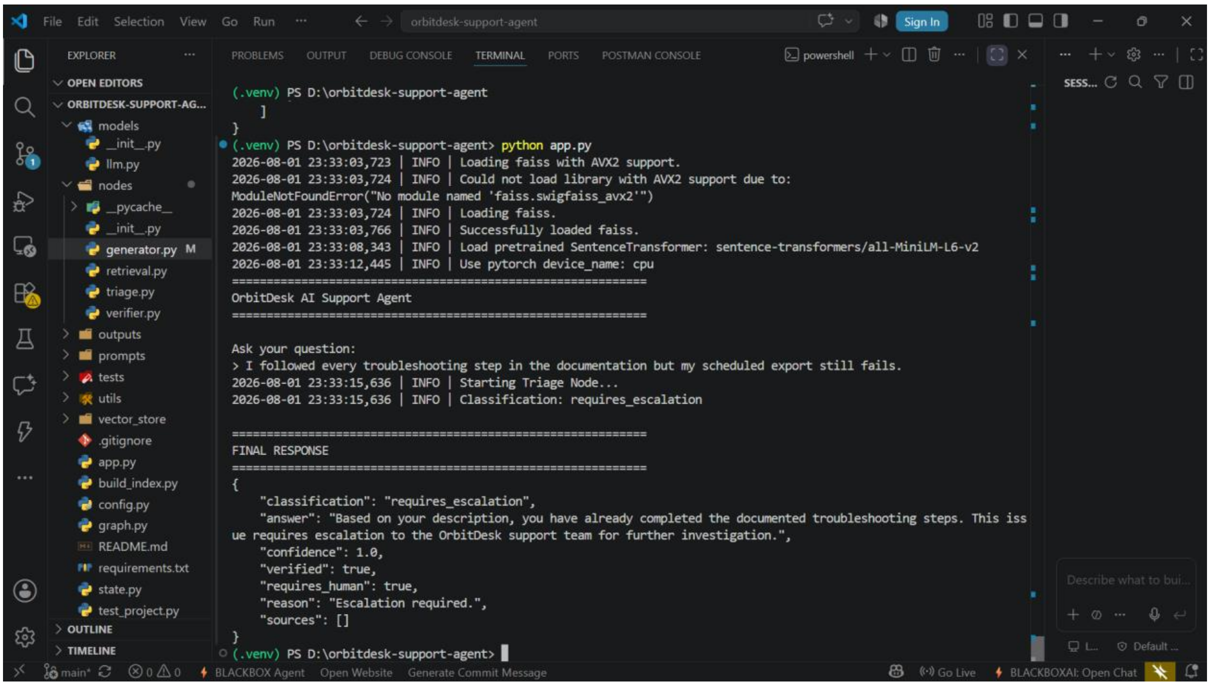


(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:56:16,882 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:56:16,883 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:56:16,883 | INFO | Loading faiss.
2026-08-01 23:56:16,925 | INFO | Successfully loaded faiss.
2026-08-01 23:56:21,489 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:56:25,302 | INFO | Use pytorch device_name: cpu
==
OrbitDesk AI Support Agent
==
Ask your question:
> Can a Viewer create API credentials?
2026-08-01 23:56:29,234 | INFO | Starting Triage Node...
2026-08-01 23:56:29,234 | INFO | Classification: answerable
2026-08-01 23:56:29,235 | INFO | Starting Retrieval Node...
Batches: 100%| | 1/1 [00:00<00:00, 19.41it/s]
2026-08-01 23:56:29,291 | INFO | Retrieved 3 documents.
========== RETRIEVAL ==========
05_api_credentials (knowledge_base)
02_roles_and_permissions (knowledge_base)
CASE-1058 (resolved_case)
==============================
2026-08-01 23:56:29,295 | INFO | Starting Generator Node...
2026-08-01 23:56:29,295 | INFO | Answer Generated Successfully.
========== GENERATOR ==========
No. Viewers cannot create workspace API credentials. Only Owners and Admins can create or revoke workspace API credentials.
==============================
2026-08-01 23:56:29,298 | INFO | Starting Verification Node...
2026-08-01 23:56:29,298 | INFO | Verification Passed.
==
FINAL RESPONSE
==
{
"classification": "answerable",
"answer": "No. Viewers cannot create workspace API credentials. Only Owners and Admins can create or revoke workspace API credentials.",
"confidence": 0.9,
"verified": true,
"requires_human": false,
"reason": "Answer verified.",
"sources": [
{
"source_id": "05_api_credentials",
"type": "knowledge_base"
},
{
"source_id": "02_roles_and_permissions",
"type": "knowledge_base"
}
]
}
(.venv) PS D:\orbitdesk-support-agent>

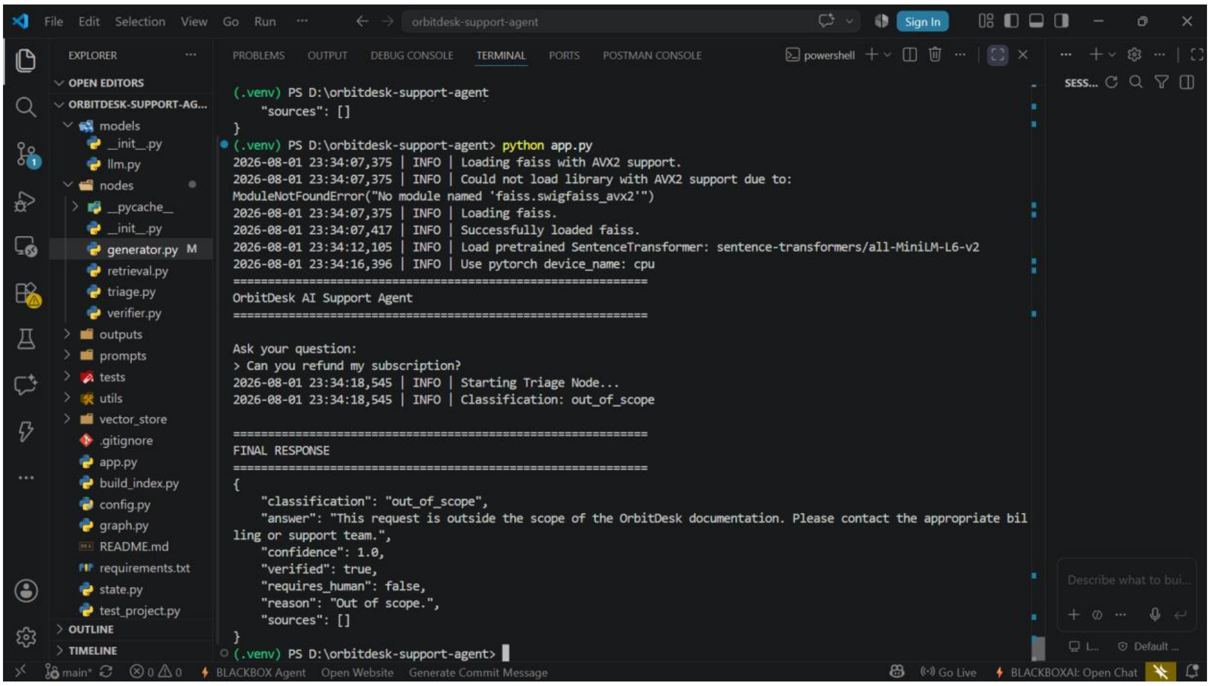


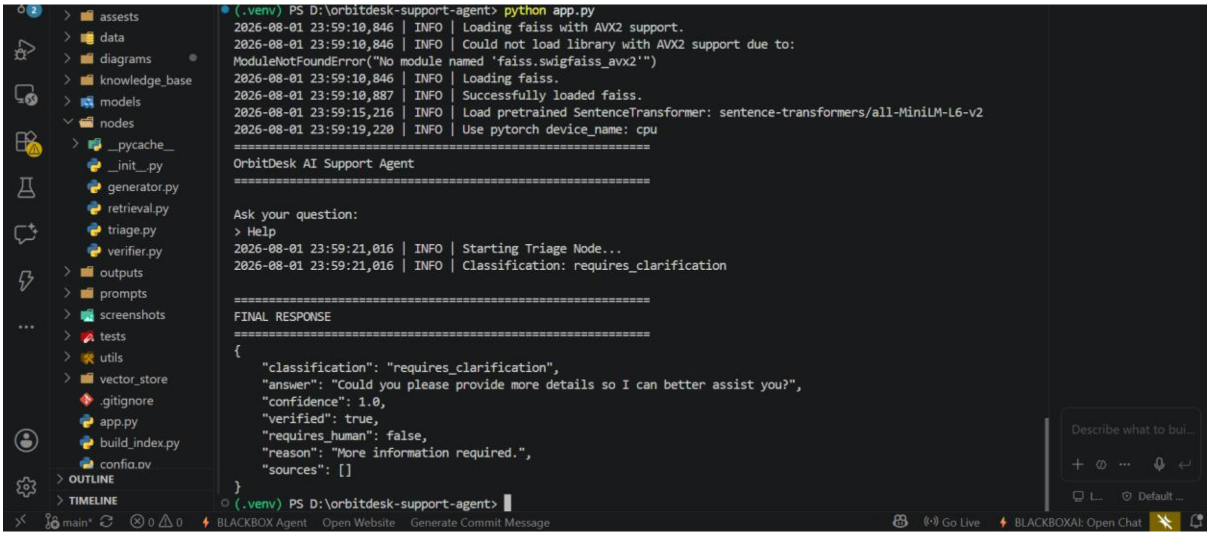
orbitdesk-support-agent
EXPLORER
OPEN EDITORS
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:57:51,915 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:57:51,916 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:57:51,916 | INFO | Loading faiss.
2026-08-01 23:57:51,958 | INFO | Successfully loaded faiss.
2026-08-01 23:57:56,420 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:58:00,335 | INFO | Use pytorch device_name: cpu
OrbitDesk AI Support Agent
Ask your question:
> What should I do if my API credential secret is exposed?
2026-08-01 23:58:03,331 | INFO | Starting Triage Node...
2026-08-01 23:58:03,332 | INFO | Classification: answerable
2026-08-01 23:58:03,332 | INFO | Starting Retrieval Node...
Batches: 100%| | 1/1 [00:00<00:00, 34.48it/s]
2026-08-01 23:58:03,365 | INFO | Retrieved 3 documents.
========== RETRIEVAL ==========
05_api_credentials (knowledge_base)
10_security_and_safe_responses (knowledge_base)
CASE-1058 (resolved_case)
2026-08-01 23:58:03,367 | INFO | Starting Generator Node...
2026-08-01 23:58:03,367 | INFO | Answer Generated Successfully.
========== GENERATOR ==========
If an API credential secret is exposed, revoke the credential immediately and create a replacement. Credential secrets cannot be recovered.
2026-08-01 23:58:03,368 | INFO | Starting Verification Node...
2026-08-01 23:58:03,368 | INFO | Verification Passed.
FINAL RESPONSE
{
"classification": "answerable",
"answer": "If an API credential secret is exposed, revoke the credential immediately and create a replacement. Credential secrets cannot be recovered.",
"confidence": 0.9,
"verified": true,
"requires_human": false,
"reason": "Answer verified.",
"sources": [
{
"source_id": "05_api_credentials",
"type": "knowledge_base"
},
{
"source_id": "10_security_and_safe_responses",
"type": "knowledge_base"
}
]
}
(.venv) PS D:\orbitdesk-support-agent>

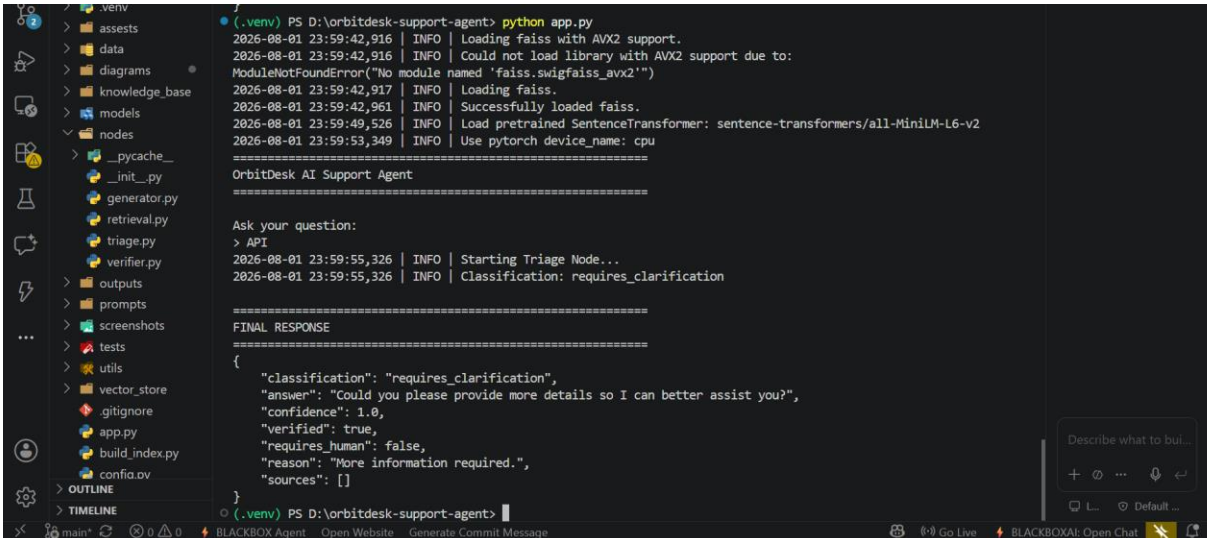

(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:53:23,608 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:53:23,608 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:53:23,608 | INFO | Loading faiss.
2026-08-01 23:53:23,653 | INFO | Successfully loaded faiss.
2026-08-01 23:53:28,143 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:53:31,975 | INFO | Use pytorch device_name: cpu
OrbitDesk AI Support Agent
Ask your question:
> What permissions does an Admin have?
2026-08-01 23:53:42,707 | INFO | Starting Triage Node...
2026-08-01 23:53:42,707 | INFO | Classification: answerable
2026-08-01 23:53:42,708 | INFO | Starting Retrieval Node...
Batches: 100%| | 1/1 [00:00<00:00, 23.52it/s]
2026-08-01 23:53:42,756 | INFO | Retrieved 3 documents.
========= RETRIEVAL =========
02_roles_and_permissions (knowledge_base)
10_security_and_safe_responses (knowledge_base)
CASE-1058 (resolved_case)
2026-08-01 23:53:42,759 | INFO | Starting Generator Node...
2026-08-01 23:53:42,759 | INFO | Answer Generated Successfully.
========= GENERATOR =========
Admins can manage members, workspace settings, connections, and create or revoke workspace API credentials.
2026-08-01 23:53:42,760 | INFO | Starting Verification Node...
2026-08-01 23:53:42,760 | INFO | Verification Passed.
FINAL RESPONSE
{
"classification": "answerable",
"answer": "Admins can manage members, workspace settings, connections, and create or revoke workspace API credentials.",
"confidence": 0.9,
"verified": true,
"requires_human": false,
"reason": "Answer verified.",
"sources": [
{
"source_id": "02_roles_and_permissions",
"type": "knowledge_base"
},
{
"source_id": "10_security_and_safe_responses",
"type": "knowledge_base"
}
]
}
(.venv) PS D:\orbitdesk-support-agent>



File Edit Selection View Go Run
orbitdesk-support-agent
Sign In
EXPLORER
OPEN EDITORS
ORBITDESK-SUPPORT-AG...
models
__init__.py
llm.py
nodes
__pycache__
__init__.py
generator.py M
retrieval.py
triage.py
verifier.py
outputs
prompts
tests
utils
vector_store
.gitignore
app.py
build_index.py
config.py
graph.py
README.md
requirements.txt
state.py
test_project.py
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
powershell
(.venv) PS D:\orbitdesk-support-agent
]
}
(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:33:03,723 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:33:03,724 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:33:03,724 | INFO | Loading faiss.
2026-08-01 23:33:03,766 | INFO | Successfully loaded faiss.
2026-08-01 23:33:08,343 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:33:12,445 | INFO | Use pytorch device_name: cpu
==
OrbitDesk AI Support Agent
==
Ask your question:
> I followed every troubleshooting step in the documentation but my scheduled export still fails.
2026-08-01 23:33:15,636 | INFO | Starting Triage Node...
2026-08-01 23:33:15,636 | INFO | Classification: requires_escalation
==
FINAL RESPONSE
==
{
"classification": "requires_escalation",
"answer": "Based on your description, you have already completed the documented troubleshooting steps. This iss
ue requires escalation to the OrbitDesk support team for further investigation.",
"confidence": 1.0,
"verified": true,
"requires_human": true,
"reason": "Escalation required.",
"sources": []
}
(.venv) PS D:\orbitdesk-support-agent>
SESS...
Describe what to bui...
Default ...
main* 0 0
BLACKBOX Agent Open Website Generate Commit Message
Go Live
BLACKBOXAI: Open Chat


(.venv) PS D:\orbitdesk-support-agent
"sources": []
}
(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:34:07,375 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:34:07,375 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:34:07,375 | INFO | Loading faiss.
2026-08-01 23:34:07,417 | INFO | Successfully loaded faiss.
2026-08-01 23:34:12,105 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:34:16,396 | INFO | Use pytorch device_name: cpu
==
OrbitDesk AI Support Agent
==
Ask your question:
> Can you refund my subscription?
2026-08-01 23:34:18,545 | INFO | Starting Triage Node...
2026-08-01 23:34:18,545 | INFO | Classification: out_of_scope
==
FINAL RESPONSE
==
{
"classification": "out_of_scope",
"answer": "This request is outside the scope of the OrbitDesk documentation. Please contact the appropriate billing or support team.",
"confidence": 1.0,
"verified": true,
"requires_human": false,
"reason": "Out of scope.",
"sources": []
}
(.venv) PS D:\orbitdesk-support-agent>


(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:59:10,846 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:59:10,846 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:59:10,846 | INFO | Loading faiss.
2026-08-01 23:59:10,887 | INFO | Successfully loaded faiss.
2026-08-01 23:59:15,216 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:59:19,220 | INFO | Use pytorch device_name: cpu
==
OrbitDesk AI Support Agent
==
Ask your question:
> Help
2026-08-01 23:59:21,016 | INFO | Starting Triage Node...
2026-08-01 23:59:21,016 | INFO | Classification: requires_clarification
==
FINAL RESPONSE
==
{
"classification": "requires_clarification",
"answer": "Could you please provide more details so I can better assist you?",
"confidence": 1.0,
"verified": true,
"requires_human": false,
"reason": "More information required.",
"sources": []
}
(.venv) PS D:\orbitdesk-support-agent>


.venv
assests
data
diagrams
knowledge_base
models
nodes
__pycache__
__init__.py
generator.py
retrieval.py
triage.py
verifier.py
outputs
prompts
screenshots
tests
utils
vector_store
.gitignore
app.py
build_index.py
config.py
OUTLINE
TIMELINE
(.venv) PS D:\orbitdesk-support-agent> python app.py
2026-08-01 23:59:42,916 | INFO | Loading faiss with AVX2 support.
2026-08-01 23:59:42,916 | INFO | Could not load library with AVX2 support due to:
ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")
2026-08-01 23:59:42,917 | INFO | Loading faiss.
2026-08-01 23:59:42,961 | INFO | Successfully loaded faiss.
2026-08-01 23:59:49,526 | INFO | Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2
2026-08-01 23:59:53,349 | INFO | Use pytorch device_name: cpu
==
OrbitDesk AI Support Agent
==
Ask your question:
> API
2026-08-01 23:59:55,326 | INFO | Starting Triage Node...
2026-08-01 23:59:55,326 | INFO | Classification: requires_clarification
==
FINAL RESPONSE
==
{
"classification": "requires_clarification",
"answer": "Could you please provide more details so I can better assist you?",
"confidence": 1.0,
"verified": true,
"requires_human": false,
"reason": "More information required.",
"sources": []
}
(.venv) PS D:\orbitdesk-support-agent>
Describe what to bui...
main*
BLACKBOX Agent
Open Website
Generate Commit Message
Go Live
BLACKBOXAI: Open Chat